

Wk2.1: Ch2 Stat learning framework

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Recap: $Y = f(X) + \epsilon$

$X = (X_1, \dots, X_p)$ indep

$Y = \text{dep variable}$

Approximate $f(x) \approx \hat{f}(x)$ by fitting to training data

known $\hat{y} = \hat{f}(x)$

Theorem (red-irred error) Assume $f, \hat{f}$ fixed & ϵ is a random variable

$E[Y - \hat{Y}] = \underbrace{(f(x) - \hat{f}(x))}_{\text{Reducible error}} + \underbrace{Var(\epsilon)}_{\text{Irreducible error}}$

Key takeaway: $\hat{f} \approx f$ has limits by $X-Y$ relationship (ϵ is out of control)

Proof

Use linearity of $E\{\cdot\}$.

① $E\{cX\} = c E\{X\}$

② $E\{X_1 + X_2\} = E\{X_1\} + E\{X_2\}$

Assume $f, \hat{f}$ are fixed & ϵ random

$E[(Y - \hat{Y})^2] = E[(f(x) + \epsilon - \hat{f}(x))^2]$

$= E[(f(x) - \hat{f}(x)) + \epsilon]^2$

$= E[(f(x) - \hat{f}(x))^2 + 2\epsilon(f(x) - \hat{f}(x)) + \epsilon^2]$

$= E[(f(x) - \hat{f}(x))^2] + E[2\epsilon(f(x) - \hat{f}(x))] + E[\epsilon^2]$

$= (f(x) - \hat{f}(x))^2 + 2(f(x) - \hat{f}(x))E[\epsilon] + E[\epsilon^2]$

$= (f(x) - \hat{f}(x))^2 + Var(\epsilon)$

Prob review: ① $P(\cdot)$ ✓

② $E\{\cdot\}$ Now.

Expectation: fair

Random variable X (lot of numbers outcomes)

PMF: $p(x) = P(X=x)$

$\hookrightarrow E_X p(x) = 1/6$

$E[X] = \sum_x x p(x)$ (avg)

$\hookrightarrow E_X$ die: $E[X] = 7/2 = 3.5 = \mu$

$E[g(x)] = \sum_x g(x) p(x)$ (distance)

$Var(X) = E[(X - \mu)^2]$ $\mu = E[X]$

$\sigma = SD(X) = \sqrt{Var(X)}$ (same units as X)

$\hookrightarrow E_X$ Die

$Var(X) = E[(X - \mu)^2] = E[X^2] - \mu^2 = \frac{7 \cdot 13}{6} - (7/2)^2 = 2.91\bar{6}$

$E[X^2] = \sum_{x=1}^6 x^2 p(x) = \frac{1}{6} \sum_{x=1}^6 x^2 = \frac{1}{6} (1^2 + 2^2 + \dots + 6^2) = \frac{1}{6} \frac{6 \cdot 7 \cdot 13}{6} = \frac{7 \cdot 13}{6} = 7.91\bar{6}$

$\sigma = \sqrt{Var(X)} = \sqrt{2.91\bar{6}} = 1.708$

Proton spread $\bar{X}_i \sim N(\mu, \sigma)$

Conf!

$\mu \pm \sigma = 3.5 \pm 1.708$

$\hookrightarrow [3.5 - 1.708, 3.5 + 1.708]$

$= [1.791, 5.208]$ (68% of X)

How to estimate $f \approx \hat{f}$?

$\hookrightarrow$ Use known data to find an optimal $\hat{f}$.

Training set or sample $\{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$

$x_i = (x_{i1}, x_{i2}, \dots, x_{ip})$ (vector)

predictors

2 groups of methods:

① Parametric methods (LR, NLS, ...)

Assumes $f(x) = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p$

$\hat{f}(x) \approx \hat{f}(x) = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \dots + \hat{\beta}_p x_p$

Parameters to best fit to train data

Major flaw

② Non-parametric methods

Let train data "shape" $\hat{f}$ without any formula.

Decision tree (sub-divide)

Inference? Hard to interpret.

$X < 2k \rightarrow S_1$

$X \geq 2k \rightarrow S_2$

$Y \approx \hat{Y} = \bar{Y}_{S_i}$

Handle & divide more

How much $\hat{f}$ changes relative to new train data

$f \approx \hat{f}$ flexibility

$y = \hat{f}_2(x) = \hat{\beta}_0 + \hat{\beta}_1 x$ (low flex)

$y = \hat{f}_3(x) = \hat{\beta}_0 + \hat{\beta}_1 x + \hat{\beta}_2 x^2$ (more flex)

out-of-sample

Overfitting vs underfitting train dataset

In practice, simulate this to decide which $\hat{f}$.

test (out-of-sample) (MSE_{te})

error

underfit

overfit

train error (MSE_{tr})

flexibility of $\hat{f}$ (DoF)

(Text discussion!)

$\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$ Bias variance - trade-off (key framework for model)

The "best" $\hat{f}$ balances:

① Predictive accuracy is best?

② Interperability (stat inference)

Theorem (No free lunch)

No best $\hat{f}$ winner! Try many & eval the best.

ML, LR, NN, ...

Evaluate Accuracy (many ways, basic)

(Train error) $MSE_{train} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$ (reg only)

(Test error) $MSE_{test} = \frac{1}{K} \sum_{i=1}^K (y_{i0} - \hat{y}_{i0})^2$

$\hookrightarrow$ Cross-validated is better way. (Ch.)

Theorem (BDSZ, bias variance trade-off)

Assume f fixed but $\hat{f}$ is random WRT X (train data)

ϵ is random. Then, for (X_0, y_0) unseen data

$E[(y_0 - \hat{y}_0)^2] = Var(\hat{f}(X_0)) + [Bias(\hat{f}(X_0))]^2 + Var(\epsilon)$

Avg test error

$\hat{f}$ variability per training data X

$\hat{f}$ bias

biased error

WRT train data

	Low flex	High flex
Small	Small	Big
Big	Big	Small

Key Idea

TBC